Branch: MCA (Data Science) Kargil	**Semester:** 2
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Subject Name: Technical Training - I Lab	**Subject Code**: 25CAP-652
Section/Group: 25MCD KAR-1	**Date of Performance:** 31-03-2026

Experiment 8

Experiment Aim:

To apply the concept of Stored Procedures in database operations in order to perform tasks like insertion, updating, deletion, and retrieval of data efficiently, securely, and in a reusable manner within the database system.

Tools used: PostgreSQL

Objectives:

- **Procedure Implementation:** To understand how to create and use stored procedures for database operations.
- **Code Reusability:** To reduce redundancy by encapsulating logic inside procedures.

- **Data Manipulation:** To perform INSERT, UPDATE, DELETE operations using procedures.
- **Data Retrieval:** To understand the use of functions for returning result sets.

Theory:

Stored Procedures are precompiled SQL code blocks stored in the database that can be executed repeatedly. They improve performance, enhance security, and promote modular programming. PostgreSQL supports procedures (for actions) and functions (for returning data).

Experiment Steps:

Step 1: Creating Table **Query:**

```
CREATE TABLE Student (
    student_id INT PRIMARY KEY,
    name VARCHAR(50),
    marks INT
);
```

Step 2: Inserting Sample Data **Query:**

```
INSERT INTO Student VALUES
(1, 'Aarav', 75),
(2, 'Diya', 82),
(3, 'Kabir', 68),
```

(4, 'Ananya', 90),

(5, 'Rohan', 77),

(6, 'Sneha', 85),

(7, 'Vivaan', 92),

(8, 'Megha', 80);

	student_id [PK] integer	name character varying (50)	marks integer
1	1	Aarav	75
2	2	Diya	82
3	3	Kabir	68
4	4	Ananya	90
5	5	Rohan	77
6	6	Sneha	85
7	7	Vivaan	92
8	8	Megha	80

Step 3: Stored Procedure for INSERT **Query:**

```
CREATE OR REPLACE PROCEDURE add_student(
    p_id INT,
    p_name VARCHAR,
    p_marks INT)
LANGUAGE plpgsql
AS $$
BEGIN
    INSERT INTO Student(student_id, name, marks)
    VALUES (p_id, p_name, p_marks);
END;
$$;
```

Step 4: Stored Procedure for UPDATE **Query:**

```
CREATE OR REPLACE PROCEDURE update_marks(
    p_id INT,
    p_marks INT
)
LANGUAGE plpgsql
AS $$
BEGIN
    UPDATE Student
    SET marks = p_marks
    WHERE student_id = p_id;
END;
$$;
```

Step 5: Stored Procedure for DELETE **Query:**

```
CREATE OR REPLACE PROCEDURE delete_student(
    p_id INT
)
LANGUAGE plpgsql
AS $$
BEGIN
    DELETE FROM Student
    WHERE student_id = p_id;
END;
$$;
```

Step 6: Function for SELECT (Retrieval)

Query:

```
CREATE OR REPLACE FUNCTION get_all_students()
RETURNS TABLE(student_id INT, name VARCHAR, marks INT)
LANGUAGE plpgsql
AS $$
BEGIN
    RETURN QUERY
    SELECT * FROM Student;
END;
$$;
```

Step 7: Calling Procedures and Function

Query:

```
-- Insert
CALL add_student(9, 'Ishaan', 88);

-- Update
CALL update_marks(2, 90);

-- Delete
CALL delete_student(3);

-- Select
SELECT * FROM get_all_students();
```

Output:

	student_id integer	name character varying	marks integer
1	1	Aarav	75
2	4	Ananya	90
3	5	Rohan	77
4	6	Sneha	85
5	7	Vivaan	92
6	8	Megha	80
7	9	Ishaan	88
8	2	Diya	90

Outcomes:

- I will be able to create and execute stored procedures.
- I will understand the difference between procedures and functions.
- I will perform CRUD operations using procedures.
- I will retrieve data using functions in PostgreSQL.